

ADHESIVES AND SEALANTS

Dorf Ketal's Tyzor® organic titanates and zirconates are versatile chemical tools that are widely used in the adhesives and sealants industry. This Technical Note describes some of the more important Tyzor® applications in adhesives and sealants, and explains how they work and how they are used.

- manufacturing catalysts for sealant/adhesive prepolymers
- adhesion promoters
- wetting agents
- dispersants
- surface protection
- water scavengers
- crosslinkers
- crosslinking catalysts
- thixotropic agents

Tyzor® 212	Tyzor® GBA	Tyzor® TE
Tyzor® 9000	Tyzor® GBO	Tyzor® TnBT
Tyzor® AA-65	Tyzor® NBZ	Tyzor® TOT
Tyzor® AA-75	Tyzor® NPZ	Tyzor® TPT
Tyzor® AA-105	Tyzor® OGT	
Tyzor® BTP	Tyzor® PITA	

Tyzor® TPT, NPT and TNBT are excellent choices for Ziegler-Natta catalysis of polybutadiene. Tyzor® TPT, TNBT, ET and BTP are suitable for esterification/polycondensation catalysis to produce polyester, polyamides and alkyd resins.

Tyzor® TPT, ET, TnBT, BTP and TOT are excellent choices for production of monomer components in sealant and adhesive binders via transesterification of methyl (meth)acrylates.

System	Tyzor® Products
butyl / synthetic rubber	TPT, TnBT, NBZ, AA-75
acrylic (<i>solvent-based</i>)	AA-75, AA-105, PITA, IBAY
acrylic (<i>water-based</i>)	TE
polysulfide	AA-75, AA-105, TE
polyurethane	TnBT, NBZ, AA-105, AA-75
silicone	TnBT, BTP, OGT, PITA, IBAY

Tyzor® titanates and zirconates are good choices for moisture-curing RTV-1 silicone sealants based on α,ω -dihydroxy polydimethyl siloxanes crosslinked with functional alkyl or vinyl silanes. Tyzor® products catalyze the crosslinking reaction of silane with silicone, either alone or in combination with tin compounds, and serve as drying agents by consuming water in the formulation to prevent premature crosslinking. Tyzor® titanates and zirconates also are efficient crosslinkers in many sealant systems that don't use silanes as bridging agents.

Benzamido and alkoxy crosslinking systems work well with Tyzor® products. Product selection depends on the silicone system and desired stability or reactivity. Tyzor® PITA or IBAY provide a good balance of stability and reactivity. Tyzor® TnBT or BTP offer higher reactivity, and Tyzor® AA-series products provide reduced reactivity.

Tyzor® titanates are usually added prior to adding the crosslinker, or occasionally with the crosslinker. Typical application rates range from 1 to 10 percent with respect to the siloxane. Tyzor® products are compatible with most common components in RTV-1 silicone

$$\text{H} \left[\begin{array}{c} \text{CH}_3 \\ | \\ \text{OSi} \\ | \\ \text{CH}_3 \end{array} \right]_n \text{OH} + \text{RSiX}_3 \xrightarrow[\text{(+ H}_2\text{O / -HX)}]{\text{Ti / Zr}}$$

$$\text{H} \left[\begin{array}{c} \text{CH}_3 \\ | \\ \text{OSi} \\ | \\ \text{CH}_3 \end{array} \right]_n \text{O} \begin{array}{c} \text{R} \\ | \\ \text{Si} \\ | \\ \text{O} \\ | \\ \text{CH}_3 - \text{Si} - \text{CH}_3 \\ | \\ \text{O} \\ | \\ \text{H} \end{array} \text{O} \left[\begin{array}{c} \text{CH}_3 \\ | \\ \text{SiO} \\ | \\ \text{CH}_3 \end{array} \right]_n \text{H}$$

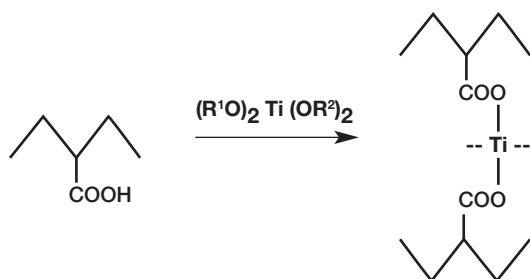
sealant formulations, including extenders and reinforcing fillers such as CaCO₃, pigments, quartz and fumed silica, as well as wetting agents and other additives.

Crosslinking Chart

Group X to be Split	System Name	Type of Crosslinking
R-NH	amino	basic
CH ₃ -COO	acetoxo	acidic
RR'C = N-O	oxime	neutral
Ar-CO-N(R)	benzamido	neutral
RO (R = CH ₃ , C ₂ H ₅)	alkoxy	neutral

Acrylics

Tyzor® titanates and zirconates are very effective crosslinkers for acrylics and other non-aqueous sealants and adhesives. They react with carboxylic groups and OH-groups in co-polymerized acrylic acid, and the alcohol stabilizer delays the crosslinking reaction. Crosslinking occurs spontaneously after the solvent evaporates. Tyzor® AA-75, AA-105, TNBT, BTP or TOT provide good stability and adhesion. Application rates range from 0.3 to 1.5 percent based on the acrylic polymer.



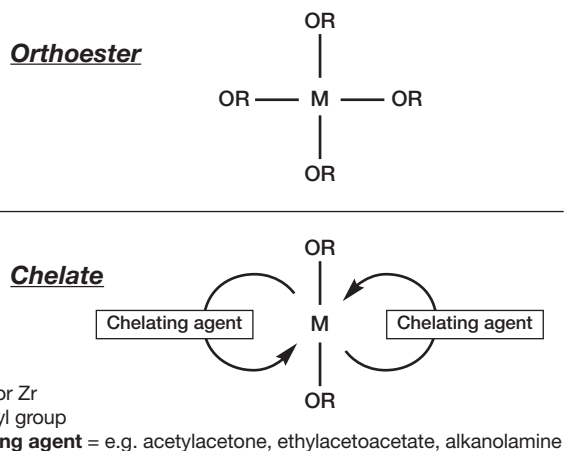
Tyzor® titanates and zirconates are effective in many other adhesive and sealant systems. A partial list follows.

Tyzor® Products for Sealants and Adhesives

System	Tyzor® Products
acrylic (<i>aqueous</i>)	TE
polyurethane (PUR)	AA-75, AA-105, TnBT
epoxy (EP)	AA-75, AA-105, TnBT
butyl rubber, isoprene, chloroprene	TnBT, NBZ, AA-75

TYZOR® CHEMISTRY

Tyzor® titanates and zirconates include orthoesters and chelates with the following general formulas:



Tyzor® **orthoesters** are very reactive compounds that react with water to produce titanium oxide hydrate – or titanium dioxide if heated. They also react with inorganic substrates or functional organic polymers (OH, COOH, NH₂, CONH and similar groups) to produce stable chelates.

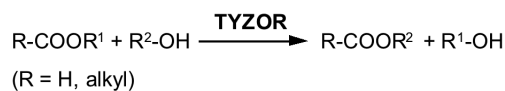
Tyzor® **chelates** include complexing agents such as acetylacetone, ethylacetoacetate, glycol ethers or alkanolamines to reduce water reactivity. Tyzor® triethanolamine chelates are very stable in water.

TYZOR® APPLICATIONS

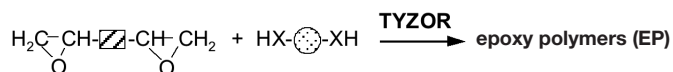
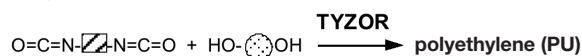
Manufacturing Catalysts

Tyzor® titanates and zirconates act as Lewis Acids that catalyze reactions such as polymerizations, polycondensations, polyadditions, esterifications and transesterifications.

Esterification / Transesterification



Polyaddition



They can be used as catalysts or additives. Reaction conditions are mild and easy to control, yields are high and waste is low.

Effects: Tyzor® products eliminate side reactions via alcohol dehydration. They produce low color and high yields.

Use: They are used to manufacture plasticizers, acrylates, methacrylates, polyamides, polyester, polyurethanes, polyepoxides (epoxys) and related compounds. Typical application rates range from 0.01 to 0.5 percent.

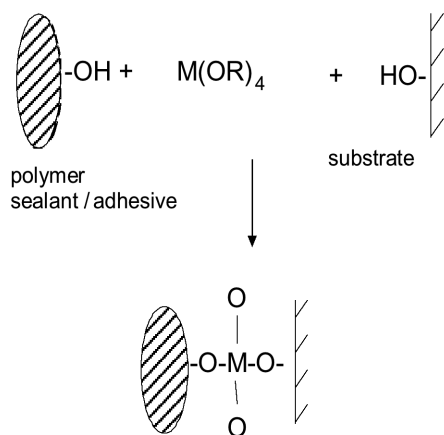
Adhesion Promoters

Highly reactive Tyzor® titanates and zirconates dramatically improve sealant and adhesive adhesion on a wide range of surfaces, including reactive and unreactive inorganic materials and even organic polymers. Tyzor® products can be used as additives at rates ranging from 0.5 to 3 percent or as primers in solution with alcohols such as isopropanol at rates ranging from 0.5 to 5 percent. Synergistic effects can be achieved when used with organo-silanes.

Effects: Adhesion and moisture resistance are improved, along with resistance to chemicals and alkalis, and electrical properties also improve.

Use: Tyzor® products are often applied as primer solutions on inorganic substrates such as glass, mineral building material or metal, and they also can be applied as polymer additives. They can be applied to plastics pretreated with corona, plasma or flame, and they can be used in combination with organofunctional silanes.

Mechanism of Adhesion Promotion



Wetting Agents

Hydrophilic inorganic or organic substrates can be made more oleophilic with a thin titanate or zirconate coating.

Effects: Tyzor® products reduce surface tension, improving wetting by organic resins.

Use: Tyzor® titanates and zirconates can be applied as primers or used as polymer additives.

Dispersants

Tyzor® titanates and zirconates can be used to modify the surfaces of pigments and fillers, improving polymer compatibility.

Effects: Pigment and filler dispersion improve and viscosity declines, allowing filler content to be increased. Modified materials have improved resistance to moisture and chemicals.

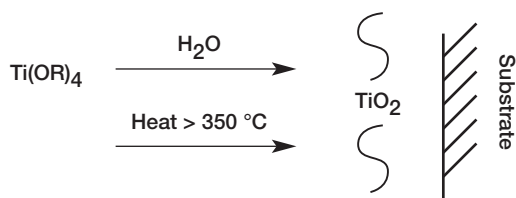
Use: Tyzor® products can be used as polymer additives or as pretreatments for fillers or pigments in highly filled sealants.

Surface Protection

A protective metal oxide layer can be formed on organic or inorganic substrates by allowing a thin coating of Tyzor® titanates or zirconates to hydrolyze in air.

Effect: The protective oxide layer improves electrical properties as well as resistance to water, chemicals, alkali and heat, and improves adhesion.

Use: Substrates can be primed with titanate/zirconate solutions. Alternatively, the coatings can be applied by partial hydrolysis and heat [$> 120 - 140\text{ }^{\circ}\text{C}$ ($248 - 284\text{ }^{\circ}\text{F}$)] or by vapor deposition at temperatures exceeding $350\text{ }^{\circ}\text{C}$ ($662\text{ }^{\circ}\text{F}$).



Water Scavengers

Water reactivity of selected Tyzor® titanates/zirconates can be exploited to remove unwanted water by forming polymeric titanium oxide hydrate or titanium dioxide and the corresponding alcohol. The resulting solid hydrous TiO_2 can be removed by filtration.

Effect: Tyzor® products act as water scavengers or drying agents, preventing pre-reactions caused by moisture and increasing storage stability of sealants and adhesives.

Use: Tyzor® titanates and zirconates are typically added to silicone and PU sealants at rates ranging from 0.5 to 5 percent.

Crosslinkers

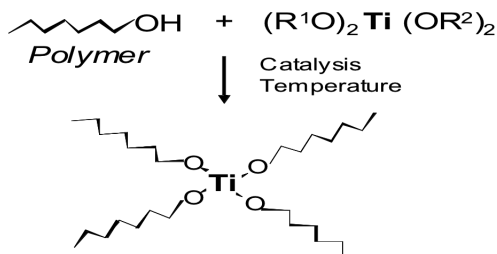
Tyzor® titanates and zirconates react with OH or COOH polymer groups to produce crosslinked polymers with higher molecular weights. Tyzor® chelates make it relatively easy to control reactivity, preventing premature crosslinking and improving stability in storage.

Effects: Tyzor® products are effective curatives and gelling agents, increase drying rates, and improve resistance to water, chemicals and heat.

Use: Tyzor® products are typically applied as additives in silicones, acrylates, glues and similar formulations.

Crosslinking Catalysts

Tyzor® titanates and zirconates are effective crosslinking catalysts that provide excellent reaction control and also improve sealant and adhesive performance.



Effect: Tyzor® products increase reaction rates in the organofunctional silanes used in silicone sealants, and they also are used in crosslinking/addition reactions to produce polyurethanes or epoxy systems.

Use: Tyzor® additives are high-performance, non-toxic replacements for toxic tin compounds.

Thixotropic Agents

Water-stable Tyzor® titanium and zirconium chelates undergo reversible crosslinking reactions via hydrogen bridge bonds to form gel structures with suitable polymers or encapsulated polymer particles.

Effect: These gels are reversibly deformed by shear and will rapidly regain their high initial viscosity when shear forces are removed (thixotropic behavior).

Use: Tyzor® titanates and zirconates are typically added with polysaccharide-encapsulated polymer particles or as the final ingredient in water-based acrylics. Application rates typically range from 0.5 to 5 percent.

Moisture Sensitivity

Tyzor® titanate and zirconate **orthoesters** such as Tyzor® TPT react rapidly with water, so care must be taken to exclude moisture from containers, blending equipment and transfer lines when these products are used.

Tyzor® **chelates** are less sensitive to water and moisture, and most can be used in aqueous solution.



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